

AstroSuite Pro

User Manual | Version 1.0.4

A complete astrophotography workflow: calibration and stacking through the bundled Siril engine, then a full image-processing suite built into its own editor. Previously released as AstroStacker Pro.

Chapter	What it covers
1. Getting around	The menu bar, toolbar, panels and views
2. Loading your frames	Lights, darks, flats, bias and smart import
3. Checking your frames	The metrics columns and what they mean
4. Stacking	Core parameters, advanced settings, timing
5. The Image Editor	Zoom, compare, transforms, Mini Viewers
6. How editing works	Iterative tools, Undo, History
7. The processing tools	Every tool, in its own group
8. Working with narrowband	A start-to-finish walkthrough
9. Masks	Protecting part of an image from a tool
10. Plate solving and colour	Image Solver, SPCC, catalogues
11. Troubleshooting	What to check when something fails
12. Credits and licence	Who wrote what, and the terms

This manual is generated from the application itself - every tool description below is the same text the app shows in its Information Panel, so the two cannot drift apart.

1. Getting around

The window is arranged in bands, top to bottom: the menu bar, the icon toolbar, a toolbar for whichever view is showing, the view itself, and the Information Panel with the Console.

The menu bar

Menu	What is in it
File	Open an image, add Lights / Darks / Flats / Bias, Smart Import, Save Image, Save Image As, Open Output Folder, Clear Frame List
Edit	Undo, Redo, Refresh Metrics, Clear Console, and hiding either panel
Tools	Subframe Selector, Blink Tool, Load Stacking Profile, and every processing tool grouped as in the sidebar
Settings	Hover Tooltips, Auto-Open Result in Editor, Advanced Settings (hover it for a specific tab), Clear Temp Files
Help	How Editing Works, What's New, Check for Updates
About	Version, what it is built on, licensing and credits

The icon toolbar

One row of icons for the things used most. Hover any of them and its name appears under the pointer, with a fuller description in the Information Panel.

- **Open, Save** - an image into the Editor, or the edited image back out.
- **Analyze Frames** - registers and measures every frame to fill in the metrics columns, without stacking.
- **Undo, Redo, Refresh Metrics.**
- **Advanced Settings, Output Folder.**
- **Flip H, Flip V, Rotate 90 left / right** - lossless, nothing is interpolated.
- **Info + Console** - hides both, giving their height to the image.
- **View FITS Header, Image Solver.**
- **STF Stretch, Linked, Unlinked, Debayer** - how the image is DISPLAYED. A green outline means active.
- **Crop.**
- **Files Viewer / Image Editor** - which view is showing, with the auto-open toggle between them.

The left panel

Two vertical tabs on the window's edge choose what it shows, and clicking the tab that is already showing hides the panel altogether - handing its width to the image. Everything on it is also reachable from the menus, so nothing is lost while it is hidden.

- **STACKING** - frame types, core parameters, output settings, profiles.
- **HISTORY** - every tool applied to the current image, in order. Click any step to go back to how the image looked at that point. Earlier sessions on the same image are kept permanently, so months later you can see what you did and when.

F9 hides and shows the panel; F8 does the same for the Information Panel and Console together.

2. Loading your frames

Add frames from the STACKING panel, the File menu, or by dragging files and folders straight onto the frame list.

Frame type	What it is for
Lights	Your actual exposures of the target.
Darks	Same exposure, same temperature, lens capped - removes sensor noise and hot pixels.
Flats	An evenly lit field - removes dust shadows and vignetting.
Bias	The shortest exposure the camera allows - the sensor's read offset.

Smart Import Files and **Smart Import Folder** read each file's header and sort it into the right category for you. Worth using when a folder holds a mixture.

Masters instead of raw calibration frames

If you already have a master dark, flat or bias, add it as that frame type and it is recognised as already stacked - it will be used directly rather than combined again.

Mixed exposures and filters

Frames are grouped by filter, exposure length and sensor temperature, so each group is calibrated with the matching master dark. Mixed exposures under one filter are calibrated separately and then merged before registration, which is correct - dark current depends on both exposure and temperature.

3. Checking your frames

Analyze Frames registers and measures every frame without stacking, filling in the columns below. Hover any column header in the app for the same description.

Column	What it tells you
Stars	How many stars were detected. Far fewer than its neighbours usually means cloud, haze or a focus problem. The count caps at Max Stars, so a run of identical values is the cap.
FWHM	Average apparent star size in pixels - the best single measure of sharpness. LOWER IS BETTER.
Roundness	How circular the stars are, 0 to 1. HIGHER IS BETTER. Low means elongated: wind, mount problems, or field rotation.
BG Noise	Measured noise in the background sky. LOWER IS BETTER.
Background	Overall sky brightness. Not good or bad in itself, but a frame much brighter than its neighbours was probably shot through cloud or twilight.
RMSE	How closely this frame's stars matched the reference after registration, in pixels. LOWER IS BETTER.
Sat %	Percentage of fully saturated pixels. Some is normal on bright stars; a sharp rise suggests over-exposure or a satellite.

Untick a frame to exclude it from the stack - it stays in the list, so you can change your mind. The Subframe Selector plots these metrics if you would rather judge them graphically, and the Blink Tool steps through the frames visually.

4. Stacking

Set the Core Parameters on the STACKING panel, then press START STACKING on the frame list row.

Core parameters

Setting	What to know
Star Detection Threshold	How readily a star is detected. Lower finds more, including noise.
Light Stacking Algorithm	Winsorized Clipping is a good default. All five of Siril's methods and all eight rejection types are available.
Sigma Low / High	How aggressively outliers are rejected. 3 / 3 suits most sets; lower rejects more.
Framing	Standard, Mosaic, Auto-Crop or Centre of Gravity - how the output frame is sized against the registered stack.
Image Weighting	FWHM suits uniform exposures. Background Noise is the better choice for mixed exposure lengths, since it weights by measured signal rather than sharpness.
Output Format	16-bit TIFF, 32-bit TIFF or 32-bit FITS.

Advanced Settings

Calibration, alignment, rejection, drizzle, frame filtering and output options, on tabs. Settings, then hover Advanced Settings to open it on a particular tab.

The timing summary

Every run ends with a breakdown - each step with its duration, frame count and share of the total. If a run feels slow, compare it against a previous one rather than guessing. Registration and calibration are normally the two largest steps.

5. The Image Editor

Double-click a file in the Files Viewer, use Open, or let a finished stack open itself if Auto-Open Result is on.

Control	What it does
Mini Viewer	Opens the image in its own window - useful for comparing two images, and as a source for tools that combine several.
Swap	Exchanges the Editor's image with a Mini Viewer's.
Before / After	Shows the image as it was before any editing. Click again to return. Nothing is changed either way.
Zoom In / Out	One step at a time, the same amount as a wheel notch.
Fit / 1:1	Fit the whole frame, or one image pixel per screen pixel. 1:1 is the only zoom showing the real data rather than a resampled version - use it to judge sharpness and noise.
Rename	Renames the file being displayed.

How the image is displayed

The four display toggles on the icon toolbar change only what you SEE. The data on disk is untouched.

- **STF Stretch** - an automatic screen stretch so faint detail is visible. Turn it off to see the raw linear data as it really is.
- **Linked** - one stretch across all three channels, preserving colour balance. The right choice for calibrated data.
- **Unlinked** - each channel stretched separately, which neutralises a colour cast. Useful on raw frames, but it hides a genuine imbalance.
- **Debayer** - demosaics a raw one-shot colour frame for display only.

6. How editing works

GHS, Simple Stretch, Curves and the other parametric tools work in iterations. Each Apply commits straight to the working image - no save prompt interrupting you - and the tool's own settings reset, so you can build a stretch across several small steps rather than one large one.

Nothing is written to disk until you choose to save. Use Save Image or Save Image As when you are ready, rather than deciding after every adjustment.

Undo, History and Compare

- **Undo** steps back through what you have applied, and Redo brings it back.
- **History** lists every step. Click any one to return to how the image looked at that point - the steps you go back past move to redo, so nothing is lost.
- **Before / After** shows the image as it was before any editing at all, so you can check the work has actually helped.

History also keeps a permanent record per image, grouped by day. Open an image months later and you can see which tools you ran and when.

7. The processing tools

Every tool in the Process Tools sidebar, in its own group. These descriptions come from the application itself.

Masking (1)

Tool	What it does
RangeSelection Mask	Generates a mask from a range of pixel values, per PixInsight's own RangeSelection tool - Lower/Upper limit, Fuzziness (soft transitions), Smoothness (blur), Screening (proportional rather than flat white), Lightness (CIE L* vs per-channel RGB/K), Invert. Check 'Active' to apply it to the live image: white areas get fully processed by whatever tool runs next, black areas stay completely untouched - works with any tool in the app, not just this one. This window must stay open for the mask to stay active.

Image Prep (5)

Tool	What it does
View FITS Header	Shows the raw FITS header (exposure, temperature, filter, WCS, etc.) for the file currently being previewed.
Bin/Downscale	Shrinks the image by an exact integer factor (true pixel binning - averages each block down to one pixel, not a blurry resize) or a custom percentage. Useful for making a smaller version of a finished image to share online. Applies directly, same as GHS/Curves - use Undo to revert.
Set Preview Region	Optional: mark a region to keep at TRUE full resolution while adjusting sliders in GHS or Curves, while the rest of the image stays on the fast downsampled preview - like Siril's own ROI feature. Off by default and doesn't replace the normal preview.
Crop Tool	Draw a crop box on the image, then drag its edges to adjust, and click Apply Crop when ready. If any Mini Viewer windows are open, you'll be offered the same crop for those too.
Blemish Blaster	Franklin Marek's Blemish Blaster technique - drag on the image to mark a spot (hot pixel, dust donut, satellite trail fragment) for repair. Repairs by sampling 3 nearby regions that brightness-match the spot's own surroundings, avoiding other stars or gaps rather than just blurring the neighbourhood. Feather and Opacity sliders control how strong and how soft the repair is.

Calibration & Astrometry (3)

Tool	What it does
Image Solver	Runs just Siril's plate solve on its own, embedding WCS astrometry (RA/Dec, pixel scale, rotation) into the file's header - no colour calibration afterward, unlike SPCC below. Writes straight back to the same file, since only header metadata changes. Requires an internet connection - run this after cropping to avoid losing the plate solve to the crop.
SPCC	Runs Siril's plate solve + Spectrophotometric Color Calibration on the currently previewed image. Requires an internet connection. Run this after cropping to avoid losing the plate solve to the crop.
LinearFit	Fits a robust straight line matching the currently-open image's pixel values to a separate reference image's, per channel, then applies that line to the whole image. Useful for matching brightness/background levels across separately-processed masters (e.g. different narrowband filters) before combining them. Images must be pixel-for-pixel registered.

Background Tools (3)

Tool	What it does
GraXpert	Runs GraXpert's AI Background Extraction or Denoising on the currently previewed image. Requires the GraXpert path to be set in Advanced Settings -> AI Tools.
Background Extraction	Runs Siril's background extraction (subsky -rbf) on the currently previewed image. Useful for applying it AFTER other processing like NoiseXTerminator instead of only automatically before stacking, since running noise reduction on already-extracted data can give patchy results.
Background Neutralization	Equalizes the average red, green and blue background levels to a neutral grey, removing any colour cast in the sky background. A neutral background is a precondition for correct colour calibration, so this normally runs before SPCC or other colour calibration steps. Draw a Preview Region over open sky first to sample just that area, or leave it unset to use the whole image.

Noise & Sharpening (1)

Tool	What it does
Xterminator Tools	Opens BlurXTerminator, StarXTerminator, and NoiseXTerminator in one tabbed window, each running on the currently previewed image via the RC-Astro CLI. Requires the CLI path to be set in Advanced Settings -> AI Tools.

Star Tools (5)

Tool	What it does
RGB Stars to NB	Puts REAL star colour, measured from BROADBAND data, onto a narrowband image - the complement of NB to RGB Stars, which synthesises star colour when you have no broadband data at all. Needs three registered, stretched images: your RGB frame with stars (an unfiltered OSC frame counts, but NOT one shot through a narrowband filter - its star colour is already false), the same RGB frame starless, and your narrowband image starless. It recovers the star layer from the RGB by unscreening, screens it onto the narrowband, then does a CIELAB transfer taking lightness from one and colour from the other, which is what makes it more than a screen blend. Everywhere there are no stars the narrowband image is preserved exactly. Ported from Bill Blanshan's RGB Stars-to-Narrowband Transfer V14.
Star Reduction	Bill Blanshan's Star Reduction PixelMath methods (Transfer/Halo/Star) - needs a starless version of your image (e.g. from StarXTerminator) alongside the original. Select both images to begin.
Screen Stars	Recombines a starless image with a separately-processed stars-only image via the Screen blend - Bill Blanshan & Mike Cranfield's technique, the exact inverse of the \"unscreen\" math used to remove stars in the first place. Select both images to begin.
Star Stretch	Franklin Marek's Star Stretch v2.1 technique - a linear-to-nonlinear stretch tuned for a stars-only image, boosting colour and removing green cast automatically. Meant to run on a combined stars image before screening it back into a starless one.
NB to RGB Stars	Franklin Marek's NBtoRGBStars v1.6 technique - combines Ha/OIII/(optional SII) stars-only images into a realistic RGB stars image, ready to screen back in. Also accepts a single OSC image shot through a dual-band filter instead of 3 separate mono ones. Optional Apply Star Stretch runs that technique on the result as the final step.

Stretch Tools (5)

Tool	What it does
------	--------------

Simple Stretch	Permanently applies the same auto-stretch algorithm used for the viewer's own display to the actual pixel data - a quick one-slider stretch. Needed before tools that expect a stretched image (e.g. Narrowband Normalization) if you haven't stretched elsewhere.
Histogram Transformation	PixInsight-style HistogramTransformation - drag the black, midtone and white handles under a live RGB histogram, with the transfer curve drawn against the identity diagonal. Auto suggests a starting black point and midtone from the image itself. Linked across channels. Live preview in the Editor. Note the black point CLIPS: everything below it is discarded permanently.
GHS	Built from scratch in this app (not scripted through Siril) from the published GHS specification (David Payne & Mike Cranfield) - Generalized Hyperbolic, Midtone Transfer, Arcsinh, Power Law, or Linear, with full control over strength, symmetry, and shadow/highlight protection, plus Colour/Lightness/Saturation modes. Live preview.
Curves	Interactive curve editor - Full RGB, Individual RGB, Luminance Only, or Saturation. Click to add a point, drag to move it, right-click to remove it. Built directly in this app (not scripted through Siril).
VeraLux HyperMetric Stretch	A ported, verified version of Riccardo Paterniti's real VeraLux HyperMetric Stretch script for Siril - a linear-to-nonlinear stretch that preserves each pixel's original colour ratios rather than stretching R/G/B independently, avoiding the hue shift a plain histogram stretch can cause. Ready-to-Use (finished, export-ready look) or Scientific (raw, deterministic - a base for further manual tone mapping) modes, with sensor-specific luminance weighting and an Auto-Calculate solver.

Tone (1)

Tool	What it does
Camera RAW Editor	AstroSuite Pro's own expanded version of the original Brightness/Contrast/Sharpen tool - adds Exposure, Highlights/Shadows/Whites/Blacks, Vibrance, Clarity/Texture, and an adjustable-radius, noise-aware Sharpen, alongside the original 4 controls. Live preview. Only really works on already-stretched (non-linear) data - on linear data, where almost all the signal sits near black, most of these controls behave unpredictably or barely at all.

Structure & Detail (3)

Tool	What it does
Dust Lane Enhancer	Detects dust lanes and dark filaments by their SHAPE - the Frangi vesselness algorithm from medical blood-vessel detection (dust lanes are the same mathematical structure: elongated ridges) - then deepens only what it detects. Not a brightness stretch: a shape-based enhancement no other astro processing software has. Works best on a starless image - bright stars can produce a faint ring-like false response of their own. Use Show Detection Map to see exactly what it's found while tuning.
Dark Structure Enhance	Carlos Sonnenstein & Oriol Lehmkuhl's DarkStructureEnhance (the real PixInsight script, ported faithfully) - builds a mask of dark large-scale structure via wavelet layer decomposition, then darkens through it, deepening dark nebulosity and dust against its surroundings while stars and bright structure stay untouched. Companion to the Dust Lane Enhancer: this boosts dark structure generally, whatever its shape, vs that tool's filament-specific detection. Layers controls what counts as large-scale; Show Mask displays exactly what's selected.

Multiscale Local Contrast	Boosts contrast only at the structure SIZES you choose, leaving every other scale mathematically untouched - a genuinely different lever from a stretch or curve, which move every scale together. Splits the image into wavelet scale layers (the same verified transform Dark Structure Enhance uses), amplifies just the selected band, and rebuilds it exactly. Background Protection holds the effect back over faint sky, where boosting would otherwise amplify noise harder than real structure - measured directly on a real image, which is exactly why it defaults on. Note it boosts STARS as well, since a star is small high-contrast structure by definition (measured: bright star bodies gained +0.105 on a real frame at default settings) - best run on a starless image, or with Strength kept moderate.
---------------------------	---

Colour Tools (5)

Tool	What it does
Astro Color Mixer	Hue-targeted colour and luminance refinement, adjusting each colour band (Red/Orange/Yellow/Green/Cyan/Blue/Purple/Magenta) independently, with built-in protection for shadows, near-neutral pixels, and highlights/stars. Adapted from Pat Cosgrove's original PixInsight Astro Color Mixer script (Cosgrove's Cosmos), via Yannick Dutertre's (Cuiv the Lazy Geek) Siril port. Only really works on already-stretched (non-linear) RGB data.
Color Masks	Bill Blanshan's hue-based colour masks - isolates one colour family (Red/Yellow/Green/Cyan/Blue/Magenta) with adjustable strength and blur, so a saturation adjustment only affects that colour, leaving everything else untouched. Invert Mask is on by default.
Color Saturation	A saturation multiplier per hue (0=unchanged, positive boosts, negative cuts) at each of the 6 fixed hue positions, smoothly interpolated between them, plus a Hue Shift to rotate the whole set together. Uses fixed control points at Color Masks' same 6 hue positions rather than a fully free-form curve.
SCNR	Subtractive Chromatic Noise Reduction - corrects a colour cast from one channel (usually green) being misaligned with the other two, protecting star colour in the process. All 5 protection methods (Average/Maximum/Minimum Neutral, Maximum/Additive Mask), plus Preserve Lightness.
Correct Magenta Stars	Removes the magenta fringing chromatic aberration leaves around stars - usually worst toward the frame corners, where the optics focus red and blue at slightly different points from green so the star's edge shows the colour green is missing from. It lifts that deficient green back toward its neighbours, which is the OPPOSITE clamp to SCNR: SCNR pulls an excess channel DOWN, this fills a deficient one UP. By default it only touches stars, found with the same detector the palette tools use and grown outward to cover the fringe ring rather than just the core - so magenta nebulosity, which is real signal in narrowband and palette images, is left alone. Turn that off for a broadband image with an overall magenta cast. Luminance-preserving, so removing a fringe does not brighten the star's edge.

Channels & Palettes (5)

Tool	What it does
Split Channels	Splits the open image into separate MONO images, one per channel, each opened in a new Mini Viewer - the source is read-only and never altered. Not needed for palette remapping (PixelMath does any mapping in one expression), but it is the only way to process each channel SEPARATELY: stretch a weak OIII harder than the Ha, denoise only the noisy one, remove stars from one and not the other, then recombine with Combine LRGB or NB to RGB Stars - both of which expect separate mono masters that a one-shot-colour frame could not previously produce inside the app. On a duo-band OSC frame, Red is your Ha and Green/Blue carry the OIII. Optionally also extracts a synthetic Luminance.

PixelMath	Applies an arithmetic expression to every pixel - one expression for all three channels, or one per channel to mix or swap them. Symbols R, G, B, L (luminance), T (whole image); functions min, max, abs, sqrt, log, exp, pow, mtf, iif; and statistics med, mean, sdev, mad that collapse an image to a single number, so things like 'R - med(R) + 0.5' work as they do in PixInsight. Live preview in the Editor as you type. Expressions are parsed and whitelisted rather than executed directly, so a typo reports a plain error instead of doing anything unexpected.
Combine LRGB / Narrowband	Combines separate mono master images into one colour image - useful after stacking a mono camera or extracting individual narrowband filters. Custom mode assigns any three images to Red/Green/Blue; HOO/SHO/HSO/HOS presets map your narrowband filters automatically. An optional Luminance master can be blended in for LRGB/LSHO-style results.
Narrowband Normalization	Bill Blanshan and Mike Cranfield's Narrowband Normalization process (HOO/SHO/HSO/HOS) on the currently previewed image. Best run on a starless, background-neutralized image.
Dual-Band to SHO	Turns a stacked dual-band / OSC colour image into a gold-and-teal SHO-style LOOK. NOT the same job as Dual-Band Extraction on the stacking side, despite the similar name - that one splits your raw CFA subs into genuine Ha and OIII masters before stacking, from real measured photosites. This works on an already-stacked colour image and SYNTHESISES the look: a duo-band frame contains no sulphur at all, so this CREATES a palette rather than revealing one - which is exactly why it is labelled as such. What IS real in the data is the OIII/Ha ratio, and this amplifies how each pixel's ratio DEVIATES from the field median, since the measured spread is small enough that merely scaling channel levels only tints the frame. Brightness is never invented - every pixel keeps the exact luminance it started with, verified to 1.8e-07 - so it changes colour only. Original code, verified on three different real frames. Chains with Narrowband Normalization rather than competing: that tool BALANCES a palette already present, this one creates one.

8. Working with narrowband

A dual-band filter on a colour camera, start to finish:

- Load your lights and calibration frames and stack as normal.
- Open the master in the Editor and crop the stacking edges away.
- Extract the channels - Split Channels, or Dual-Band to SHO if you want a synthetic SHO palette from one duo-band image.
- Process each channel separately. This is the real value of narrowband: Ha and OIII want very different stretches.
- Recombine with Combine LRGB / Narrowband, choosing a palette.
- Balance with Narrowband Normalization, which brings the channels into a sensible relationship rather than leaving one dominant.
- Finish with star and colour tools as you would any image.

A note on SPCC and narrowband. Colour calibration compares each star against its catalogue spectrum. Through a 7nm filter pair, stars emit almost nothing, so the fit is noisy and the result is often rated imprecise. That is expected rather than a fault - there is no meaningful true colour to calibrate to in a narrowband palette. SPCC earns its keep on broadband data.

9. Masks

A mask protects part of an image from whatever tool you apply next. White is fully processed, black is fully protected, grey blends.

- Open RangeSelection Mask and set the range so the mask covers what you want changed.
- Tick **Active**. The mask now applies to ANY tool's result, with no further setup.
- Apply your tool as normal - it affects only the unprotected areas.
- Closing the mask dialog clears it. Minimising does not.

Hide / Unhide toggles a translucent red overlay on the main image so you can see what is protected. The mask's own preview window shows it as literal black and white instead.

10. Plate solving and colour calibration

Image Solver works out exactly where the image is pointing and writes that astrometry into its header. **SPCC** plate solves and then calibrates colour against catalogue star spectra.

If solving fails

- **No coordinates.** Siril needs a rough starting position. Both dialogs have a target box - type the object's NAME (M42, NGC 7000, Rosette Nebula) and it is looked up for you, or enter coordinates directly. The dialog tells you whether the header already has them.
- **Wrong scale.** Solving also needs focal length and pixel size. If the header lacks them, Siril falls back on its stored preferences, which may be from a different rig entirely.
- **Server unavailable.** Plate solving normally queries a remote catalogue. Advanced Settings, Maintenance offers a one-off download of the star catalogue so it works entirely on your own machine.

11. Troubleshooting

Symptom	What to check
Stacking is slower than expected	Read the timing summary at the end of the run. If Merge is large, enable Windows Developer Mode so Siril can use symlinks instead of copying gigabytes. Registration and calibration are real work and scale with frame size and count.
Plate solve or SPCC fails	Read the console. A catalogue server error is temporary and not your data. 'No target coordinates' means the header has no pointing - type the object name in the target box.
Colour calibration says imprecise	Expected on narrowband. On broadband, check the sensor and filter are set correctly, and that the image is not starless.
A tool result looks wrong	Undo, or click back through History. Nothing is written to disk until you save.
Frames all show the same star count	That is Max Stars capping the count, not a coincidence.
Saving a FITS fails	Some files carry non-standard header keywords. These are corrected automatically on the way out - if a save still fails, the console gives the reason.
The app looks factory-fresh after an update	Settings live in %APPDATA%\AstroSuite Pro. Settings from the previous name are carried across on first run.

12. Credits and licence

AstroSuite Pro is licensed under the **GNU General Public License v3**. The full source is included in every release, as the licence requires.

Calibration and stacking are performed by **Siril**, bundled unmodified with its own licence and AUTHORS file alongside it. Siril is free software - siril.org.

The tools

Many of the processing tools are faithful ports of work by others, used with permission or under compatible licences:

Author	Tools
Franklin Marek (Seti Astro)	Blemish Blaster, Star Stretch, NB to RGB Stars
Bill Blanshan	Star Reduction, Screen Stars, Color Masks, Narrowband Normalization
David Payne and Mike Cranfield	Generalised Hyperbolic Stretch
Riccardo Paterniti	VeraLux HyperMetric Stretch
Original authors	DarkStructureEnhance

The offline star catalogue is Siril's own Gaia DR3 extract, by Richard, Melis, Knagg-Baugh and Cass, under CC BY 4.0.

Copyright (C) 2026 Stewart Oliver. This program comes with ABSOLUTELY NO WARRANTY. It is free software, and you are welcome to redistribute it under the terms of the GPL v3.